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Java SE 27 & JDK 27

Class ManagementFactory

java.lang.Object
    java.lang.management.ManagementFactory

```
public class ManagementFactory
extends Object
```

The `ManagementFactory` class is a factory class for getting managed beans for the Java platform. This class consists of static methods each of which returns one or more *platform MXBeans* representing the management interface of a component of the Java virtual machine.

Platform MXBeans

A platform MXBean is a *managed bean* that conforms to the JMX Instrumentation Specification and only uses a set of basic data types. A JMX management application and the platform MBeanServer can interoperate without requiring classes for MXBean specific data types. The data types being transmitted between the JMX connector server and the connector client are open types and this allows interoperation across versions. See the specification of MXBeans for details.

Each platform MXBean is a `PlatformManagedObject` and it has a unique `ObjectName` for registration in the platform `MBeanServer` as returned by by the `getObjectName` method.

An application can access a platform MXBean in the following ways:

1. Direct access to an MXBean interface

- Get an MXBean instance by calling the `getPlatformMXBean` or `getPlatformMXBeans` method and access the MXBean locally in the running virtual machine.
- Construct an MXBean proxy instance that forwards the method calls to a given `MBeanServer` by calling the `getPlatformMXBean(MBeanServerConnection, Class)` or `getPlatformMXBeans(MBeanServerConnection, Class)` method. The `newPlatformMXBeanProxy` method can also be used to construct an MXBean proxy instance of a given `ObjectName`. A proxy is typically constructed to remotely access an MXBean of another running virtual machine.

2. Indirect access to an MXBean interface via MBeanServer

- Go through the platform `MBeanServer` to access MXBeans locally or a specific `MBeanServerConnection` to access MXBeans remotely. The

attributes and operations of an MXBean use only *JMX open types* which include basic data types, `CompositeData`, and `TabularData` defined in `OpenType`. The mapping is specified in the MXBean specification for details.

The `getPlatformManagementInterfaces` method returns all management interfaces supported in the Java virtual machine including the standard management interfaces listed in the tables below as well as the management interfaces extended by the JDK implementation.

A Java virtual machine has a single instance of the following management interfaces:

Management Interface	ObjectName
`ClassLoadingMXBean`	`java.lang:type=ClassLoading`
`MemoryMXBean`	`java.lang:type=Memory`
`ThreadMXBean`	`java.lang:type=Threading`
`RuntimeMXBean`	`java.lang:type=Runtime`
`OperatingSystemMXBean`	`java.lang:type=OperatingSystem`
`PlatformLoggingMXBean`	`java.util.logging:type=Logging`

A Java virtual machine has zero or a single instance of the following management interfaces.

Management Interface	ObjectName
`CompilationMXBean`	`java.lang:type=Compilation`

A Java virtual machine may have one or more instances of the following management interfaces.

Management Interface	ObjectName
`GarbageCollectorMXBean`	`java.lang:type=GarbageCollector,name=`*collector's name*
`MemoryManagerMXBean`	`java.lang:type=MemoryManager,name=`*manager's name*
`MemoryPoolMXBean`	`java.lang:type=MemoryPool,name=`*pool's name*
`BufferPoolMXBean`	`java.nio:type=BufferPool,name=`*pool name*

Since:

1.5

See Also:

JMX Specification,
Ways to Access Management Metrics,

MXBean

Field Summary

Fields

Modifier and Type	Field	Description
static final String	**CLASS_LOADING_MXBEAN_NAM**	String representation of the ObjectName for the ClassLoadingMXBean.
static final String	**COMPILATION_MXBEAN_NAME**	String representation of the ObjectName for the CompilationMXBean.
static final String	**GARBAGE_COLLECTOR_MXBEAN**	The domain name and the type key property in the ObjectName for a GarbageCollectorMXBean.
static final String	**MEMORY_MANAGER_MXBEAN_DO**	The domain name and the type key property in the ObjectName for a MemoryManagerMXBean.
static final String	**MEMORY_MXBEAN_NAME**	String representation of the ObjectName for the MemoryMXBean.
static final String	**MEMORY_POOL_MXBEAN_DOMAI**	The domain name and the type key property in the ObjectName for a MemoryPoolMXBean.
static final String	**OPERATING_SYSTEM_MXBEAN_**	String representation of the ObjectName for the OperatingSystemMXBean.
static final String	**RUNTIME_MXBEAN_NAME**	String representation of the ObjectName for the RuntimeMXBean.
static final String	**THREAD_MXBEAN_NAME**	String representation of the ObjectName for the ThreadMXBean.

Method Summary

All Methods	Static Methods	

Modifier and Type	Method	Description
static ClassLoadingMXBean	**getClassLoadingMXBean**()	Returns the managed bean for the class loading system of the Java virtual machine.
static CompilationMXBean	**getCompilationMXBean**()	Returns the managed bean for the compilation system of the Java virtual machine.
static List <GarbageCollectorMXB	**getGarbageCollectorMXBea**	Returns a list of GarbageCollectorMXBean objects in the Java virtual machine.
static List <MemoryManagerMXBean	**getMemoryManagerMXBeans**(	Returns a list of MemoryManagerMXBean objects in the Java virtual machine.
static MemoryMXBean	**getMemoryMXBean**()	Returns the managed bean for the memory system of the Java virtual machine.
static List <MemoryPoolMXBean>	**getMemoryPoolMXBeans**()	Returns a list of MemoryPoolMXBean objects in the Java virtual machine.
static OperatingSystemMXBea	**getOperatingSystemMXBean**	Returns the managed bean for the operating system on which the Java virtual machine is running.
static Set<Class<? extends PlatformManagedObjec	**getPlatformManagementInt**	Returns the set of Class objects, subinterface of PlatformManagedObject, representing all management interfaces

		for monitoring and managing the Java platform.
static MBeanServer	**getPlatformMBeanServer**()	Returns the platform MBeanServer.
static <T extends PlatformManagedObject> T	**getPlatformMXBean**(Class<T> mxbeanInterface)	Returns the platform MXBean implementing the given mxbeanInterface which is specified to have one single instance in the Java virtual machine.
static <T extends PlatformManagedObject> T	**getPlatformMXBean**(MBeanServerConnection connection, Class<T> mxbeanInterface)	Returns the platform MXBean proxy for mxbeanInterface which is specified to have one single instance in a Java virtual machine and the proxy will forward the method calls through the given MBeanServerConnection.
static <T extends PlatformManagedObject> List<T>	**getPlatformMXBeans**(Class<T> mxbeanInterface)	Returns the list of platform MXBeans implementing the given mxbeanInterface in the Java virtual machine.
static <T extends PlatformManagedObject> List<T>	**getPlatformMXBeans**(MBeanServerConnection connection, Class<T> mxbeanInterface)	Returns the list of the platform MXBean proxies for forwarding the method calls of the mxbeanInterface through the given MBeanServerConnection.
	getRuntimeMXBean()	

static RuntimeMXBean		Returns the managed bean for the runtime system of the Java virtual machine.
static ThreadMXBean	**getThreadMXBean**()	Returns the managed bean for the thread system of the Java virtual machine.
static <T> T	**newPlatformMXBeanProxy**(MBeanServerConnection c String mxbeanName, Class<T> mxbeanInterface	Returns a proxy for a platform MXBean interface of a given MXBean name that forwards its method calls through the given MBeanServerConnection.

Methods declared in class Object

clone, equals, finalize, getClass, hashCode, notify, notifyAll, toString, wait, wait, wait

Field Details

CLASS_LOADING_MXBEAN_NAME

```
public static final String CLASS_LOADING_MXBEAN_NAME
```

String representation of the ObjectName for the ClassLoadingMXBean.

See Also:

Constant Field Values

COMPILATION_MXBEAN_NAME

```
public static final String COMPILATION_MXBEAN_NAME
```

String representation of the ObjectName for the CompilationMXBean.

See Also:

Constant Field Values

MEMORY_MXBEAN_NAME

```
public static final String MEMORY_MXBEAN_NAME
```

String representation of the ObjectName for the MemoryMXBean.

See Also:

Constant Field Values

OPERATING_SYSTEM_MXBEAN_NAME

```
public static final String OPERATING_SYSTEM_MXBEAN_NAME
```

String representation of the ObjectName for the OperatingSystemMXBean.

See Also:

Constant Field Values

RUNTIME_MXBEAN_NAME

```
public static final String RUNTIME_MXBEAN_NAME
```

String representation of the ObjectName for the RuntimeMXBean.

See Also:

Constant Field Values

THREAD_MXBEAN_NAME

```
public static final String THREAD_MXBEAN_NAME
```

String representation of the ObjectName for the ThreadMXBean.

See Also:

Constant Field Values

GARBAGE_COLLECTOR_MXBEAN_DOMAIN_TYPE

```
public static final String GARBAGE_COLLECTOR_MXBEAN_DOMAIN_TYPE
```

The domain name and the type key property in the ObjectName for a GarbageCollectorMXBean. The unique ObjectName for a GarbageCollectorMXBean can be formed by appending this string with ",name=*collector's name*".

See Also:

Constant Field Values

MEMORY_MANAGER_MXBEAN_DOMAIN_TYPE

```
public static final String MEMORY_MANAGER_MXBEAN_DOMAIN_TYPE
```

The domain name and the type key property in the `ObjectName` for a `MemoryManagerMXBean`. The unique `ObjectName` for a `MemoryManagerMXBean` can be formed by appending this string with "`,name=`*manager's name*".

See Also:

Constant Field Values

MEMORY_POOL_MXBEAN_DOMAIN_TYPE

```
public static final String MEMORY_POOL_MXBEAN_DOMAIN_TYPE
```

The domain name and the type key property in the `ObjectName` for a `MemoryPoolMXBean`. The unique `ObjectName` for a `MemoryPoolMXBean` can be formed by appending this string with `,name=`*pool's name*.

See Also:

Constant Field Values

Method Details

getClassLoadingMXBean

```
public static ClassLoadingMXBean getClassLoadingMXBean()
```

Returns the managed bean for the class loading system of the Java virtual machine.

Returns:

a `ClassLoadingMXBean` object for the Java virtual machine.

getMemoryMXBean

```
public static MemoryMXBean getMemoryMXBean()
```

Returns the managed bean for the memory system of the Java virtual machine.

Returns:

a `MemoryMXBean` object for the Java virtual machine.

getThreadMXBean

```
public static ThreadMXBean getThreadMXBean()
```

Returns the managed bean for the thread system of the Java virtual machine.

Returns:

a `ThreadMXBean` object for the Java virtual machine.

getRuntimeMXBean

```
public static RuntimeMXBean getRuntimeMXBean()
```

Returns the managed bean for the runtime system of the Java virtual machine.

Returns:

a `RuntimeMXBean` object for the Java virtual machine.

getCompilationMXBean

```
public static CompilationMXBean getCompilationMXBean()
```

Returns the managed bean for the compilation system of the Java virtual machine. This method returns `null` if the Java virtual machine has no compilation system.

Returns:

a `CompilationMXBean` object for the Java virtual machine or `null` if the Java virtual machine has no compilation system.

getOperatingSystemMXBean

```
public static OperatingSystemMXBean getOperatingSystemMXBean()
```

Returns the managed bean for the operating system on which the Java virtual machine is running.

Returns:

an `OperatingSystemMXBean` object for the Java virtual machine.

getMemoryPoolMXBeans

```
public static List<MemoryPoolMXBean> getMemoryPoolMXBeans()
```

Returns a list of `MemoryPoolMXBean` objects in the Java virtual machine. The Java virtual machine can have one or more memory pools. It may add or remove memory pools during execution.

Returns:

a list of `MemoryPoolMXBean` objects.

getMemoryManagerMXBeans

```
public static List<MemoryManagerMXBean> getMemoryManagerMXBeans()
```

Returns a list of `MemoryManagerMXBean` objects in the Java virtual machine. The Java virtual machine can have one or more memory managers. It may add or remove memory managers during execution.

Returns:

a list of `MemoryManagerMXBean` objects.

getGarbageCollectorMXBeans

```
public
static List<GarbageCollectorMXBean> getGarbageCollectorMXBeans()
```

Returns a list of `GarbageCollectorMXBean` objects in the Java virtual machine. The Java virtual machine may have one or more `GarbageCollectorMXBean` objects. It may add or remove `GarbageCollectorMXBean` during execution.

Returns:

a list of `GarbageCollectorMXBean` objects.

getPlatformMBeanServer

```
public static MBeanServer getPlatformMBeanServer()
```

Returns the platform `MBeanServer`. On the first call to this method, it first creates the platform `MBeanServer` by calling the `MBeanServerFactory.createMBeanServer` method and registers each platform MXBean in this platform `MBeanServer` with its `ObjectName`. This method, in subsequent calls, will simply return the initially created platform `MBeanServer`.

MXBeans that get created and destroyed dynamically, for example, memory `pools` and `managers`, will automatically be registered and deregistered into the platform `MBeanServer`.

If the system property `javax.management.builder.initial` is set, the platform `MBeanServer` creation will be done by the specified `MBeanServerBuilder`.

It is recommended that this platform MBeanServer also be used to register other application managed beans besides the platform MXBeans. This will allow all MBeans to be published through the same `MBeanServer` and hence allow for easier network publishing and discovery. Name conflicts with the platform MXBeans should be avoided.

Returns:

the platform `MBeanServer`; the platform MXBeans are registered into the platform `MBeanServer` at the first time this method is called.

See Also:

`MBeanServerFactory`,
`MBeanServerFactory.createMBeanServer()`

newPlatformMXBeanProxy

```
public static <T> T newPlatformMXBeanProxy
(MBeanServerConnection connection,
 String mxbeanName,
 Class<T> mxbeanInterface)
                                    throws IOException
```

Returns a proxy for a platform MXBean interface of a given MXBean name that forwards its method calls through the given `MBeanServerConnection`.

This method is equivalent to:

```
Proxy.newProxyInstance(mxbeanInterface.getClassLoader(),
new Class[] { mxbeanInterface }, handler)
```

where `handler` is an `InvocationHandler` to which method invocations to the MXBean interface are dispatched. This `handler` converts an input parameter from an MXBean data type to its mapped open type before forwarding to the `MBeanServer` and converts a return value from an MXBean method call through the `MBeanServer` from an open type to the corresponding return type declared in the MXBean interface.

If the MXBean is a notification emitter (i.e., it implements `NotificationEmitter`), both the `mxbeanInterface` and `NotificationEmitter` will be implemented by this proxy.

Notes:

1. Using an MXBean proxy is a convenience remote access to a platform MXBean of a running virtual machine. All method calls to the MXBean proxy are forwarded to an `MBeanServerConnection` where `IOException` may be thrown when the communication problem occurs with the connector server. If thrown, `IOException` will be wrappped in `UndeclaredThrowableException`. An application remotely accessing the platform MXBeans using proxy should prepare to catch `UndeclaredThrowableException` and handle its cause as if that cause had been thrown by the `MBeanServerConnection` interface.
2. When a client application is designed to remotely access MXBeans for a running virtual machine whose version is different than the version on which the application is running, it should prepare to catch `InvalidObjectException` which is thrown when an MXBean proxy receives a name of an enum constant which is missing in the enum class loaded in the client application. If thrown, `InvalidObjectException` will be wrappped in `UndeclaredThrowableException`.
3. `MBeanServerInvocationHandler` or its `newProxyInstance` method cannot be used to create a proxy for a platform MXBean. The proxy object

created by MBeanServerInvocationHandler does not handle the properties of the platform MXBeans described in the class specification.

Type Parameters:

T - an mxbeanInterface type parameter

Parameters:

connection - the MBeanServerConnection to forward to.

mxbeanName - the name of a platform MXBean within connection to forward to. mxbeanName must be in the format of ObjectName.

mxbeanInterface - the MXBean interface to be implemented by the proxy.

Returns:

a proxy for a platform MXBean interface of a given MXBean name that forwards its method calls through the given MBeanServerConnection, or null if not exist.

Throws:

IllegalArgumentException - if

- mxbeanName is not with a valid ObjectName format, or
- the named MXBean in the connection is not a MXBean provided by the platform, or
- the named MXBean is not registered in the MBeanServerConnection, or
- the named MXBean is not an instance of the given mxbeanInterface

IOException - if a communication problem occurred when accessing the MBeanServerConnection.

getPlatformMXBean

```
public static <T extends PlatformManagedObject> T getPlatformMXBean
(Class<T> mxbeanInterface)
```

Returns the platform MXBean implementing the given mxbeanInterface which is specified to have one single instance in the Java virtual machine. This method may return null if the management interface is not implemented in the Java virtual machine (for example, a Java virtual machine with no compilation system does not implement CompilationMXBean); otherwise, this method is equivalent to calling:

```
getPlatformMXBeans(mxbeanInterface).get(0);
```

Type Parameters:

T - an mxbeanInterface type parameter

Parameters:

`mxbeanInterface` - a management interface for a platform MXBean with one single instance in the Java virtual machine if implemented.

Returns:

the platform MXBean that implements `mxbeanInterface`, or `null` if not exist.

Throws:

`IllegalArgumentException` - if `mxbeanInterface` is not a platform management interface or not a singleton platform MXBean.

Since:

1.7

getPlatformMXBeans

```
public static <T extends PlatformManagedObject>
List<T> getPlatformMXBeans(Class<T> mxbeanInterface)
```

Returns the list of platform MXBeans implementing the given `mxbeanInterface` in the Java virtual machine. The returned list may contain zero, one, or more instances. The number of instances in the returned list is defined in the specification of the given management interface. The order is undefined and there is no guarantee that the list returned is in the same order as previous invocations.

Type Parameters:

`T` - an `mxbeanInterface` type parameter

Parameters:

`mxbeanInterface` - a management interface for a platform MXBean

Returns:

the list of platform MXBeans that implement `mxbeanInterface`.

Throws:

`IllegalArgumentException` - if `mxbeanInterface` is not a platform management interface.

Since:

1.7

getPlatformMXBean

```
public static <T extends PlatformManagedObject> T getPlatformMXBean
(MBeanServerConnection connection,
 Class<T> mxbeanInterface)
                                                                throws
IOException
```

Returns the platform MXBean proxy for `mxbeanInterface` which is specified to have one single instance in a Java virtual machine and the proxy will forward the method calls through the given `MBeanServerConnection`. This method may return `null` if the management interface is not implemented in the Java virtual machine being monitored (for example, a Java virtual machine with no compilation system does not implement `CompilationMXBean`); otherwise, this method is equivalent to calling:

```
    getPlatformMXBeans(connection, mxbeanInterface).get(0);
```

Type Parameters:

`T` - an `mxbeanInterface` type parameter

Parameters:

`connection` - the `MBeanServerConnection` to forward to.

`mxbeanInterface` - a management interface for a platform MXBean with one single instance in the Java virtual machine being monitored, if implemented.

Returns:

the platform MXBean proxy for forwarding the method calls of the `mxbeanInterface` through the given `MBeanServerConnection`, or `null` if not exist.

Throws:

`IllegalArgumentException` - if `mxbeanInterface` is not a platform management interface or not a singleton platform MXBean.

`IOException` - if a communication problem occurred when accessing the `MBeanServerConnection`.

Since:

1.7

See Also:

`newPlatformMXBeanProxy(MBeanServerConnection, String, Class)`

getPlatformMXBeans

```
public static <T extends PlatformManagedObject>
List<T> getPlatformMXBeans(MBeanServerConnection connection,
                           Class<T> mxbeanInterface)
                    throws IOException
```

Returns the list of the platform MXBean proxies for forwarding the method calls of the `mxbeanInterface` through the given `MBeanServerConnection`. The returned list may contain zero, one, or more instances. The number of instances in the returned list is defined in the specification of the given

management interface. The order is undefined and there is no guarantee that the list returned is in the same order as previous invocations.

Type Parameters:

`T` - an `mxbeanInterface` type parameter

Parameters:

`connection` - the `MBeanServerConnection` to forward to.

`mxbeanInterface` - a management interface for a platform MXBean

Returns:

the list of platform MXBean proxies for forwarding the method calls of the `mxbeanInterface` through the given `MBeanServerConnection`.

Throws:

`IllegalArgumentException` - if `mxbeanInterface` is not a platform management interface.

`IOException` - if a communication problem occurred when accessing the `MBeanServerConnection`.

Since:

1.7

See Also:

`newPlatformMXBeanProxy(MBeanServerConnection, String, Class)`

getPlatformManagementInterfaces

```
public static Set<Class<? extends
PlatformManagedObject>> getPlatformManagementInterfaces()
```

Returns the set of `Class` objects, subinterface of `PlatformManagedObject`, representing all management interfaces for monitoring and managing the Java platform.

Returns:

the set of `Class` objects, subinterface of `PlatformManagedObject` representing the management interfaces for monitoring and managing the Java platform.

Since:

1.7

Report a bug or suggest an enhancement
For further API reference and developer documentation see the Java SE Documentation, which contains more detailed, developer-targeted descriptions with conceptual overviews, definitions of terms, workarounds, and working code examples. Other versions.
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